Strict Phase-Wise Plan

Gated delivery plan: each phase has deliverables and an exit gate

Rules of the plan

- **Strict gates:** a phase starts only after the previous phase's exit criteria are met and demonstrated.
- **Scope freeze:** once a phase starts, no new features enter it. New ideas go to the backlog.
- **Always demoable:** at the end of every phase from Phase 1 onward, the system runs end to end, even if basic.
- **Cut line:** Phases 0-6 are the hackathon build. Phase 7 is hardening. Phase 8 is the roadmap slide only.
- **Timeline:** 16 working days shown. Compress by merging Phases 0-1 and 6-7 if the deadline is shorter, never by skipping a gate.

Phase table

Phase	Days	Deliverables	Exit criteria (gate)
0. Foundation	1-2	Repo, CI skeleton, event schema, manifest schema, scenario list, sample data	CI green on empty tests; event and manifest schemas frozen and reviewed
1. Gateway + events	3-4	FastAPI gateway, tool-call interception, event store, simulated Procurement and one more legit agent	100% of calls logged; gateway overhead p95 under 10 ms; replay of stored events works
2. Manifests + rules	5-6	Role manifests (3 roles), rule engine, allowlist egress checks, signed manifest loader	Out-of-manifest calls flagged 100% in tests; unsigned manifest rejected; unit coverage 80%+
3. Taint + chains	7-8	Provenance tags, kill-chain rule, delegation chain in events, authority capping, honeypot resources	Injection demo detected at the sensitive access step; callee never exceeds caller scope; honeypot hit = alert
4. Baselines + stats	9-10	Per-role baseline, Markov rarity, JS divergence, EWMA, CUSUM, embedding drift, cold-start mode	New agent covered on first event; slow-drift scenario caught; false-positive rate 5% or less on benign set
5. Risk + response	11-12	Score engine with overrides, explanation text, 4 response tiers, approval webhook, token restriction	Every alert shows top 3 reasons; tiers enforce correctly; other agents unaffected during isolation
6. Dashboard + demo	13-14	Live dashboard, scenario runner, one-command demo, demo script	Scripted demo runs clean 3 times in a row from a fresh start
7. Hardening + eval	15-16	Evaluation report, evasion tests, load test, security review, docs, CI gates tuned	Recall 90%+, FPR 5% or less, detection within 3 events, all CI gates green, no critical findings
8. Roadmap	post	Kafka pipeline, framework SDKs (LangGraph, AutoGen, MCP), SIEM and IAM connectors, compliance mapping	Presented as roadmap only; not built for the hackathon

Daily rhythm

- Start of day: confirm the phase goal and today's slice. End of day: merge to main, CI green, short written status.
- Every merge goes through CI; nothing is merged with a failing gate.
- If a phase runs over by more than one day, drop its lowest-priority deliverable and move it to the backlog. Do not extend the phase.

Risk register

Risk	Likelihood	Response
Embedding model slows scoring	Medium	Run async; fall back to rules and statistics only
Too many false alarms	Medium	Shadow mode first; tune on the benign set before enabling blocking
Demo fails on the day	Low	Recorded backup video; one-command reset; tested 3 times in Phase 6
Scope creep	High	Scope freeze rule and cut line above
Evaluation numbers weak	Medium	Start the labelled scenario set in Phase 0 and grow it every phase